

11.1 部分习题参考解答

1. (1) $f'_x = f'_y = \sin(x+y) + (x+y)\cos(x+y)$;

(2) $f'_x = 4x \tan(x^2 + y^2) \sec^2(x^2 + y^2)$, $f'_y = 4y \tan(x^2 + y^2) \sec^2(x^2 + y^2)$;

(3) $f'_x = \frac{xy \ln xy - (1+xy) \ln(1+xy)}{x(1+xy) \ln^2 xy}$, $f'_y = \frac{xy \ln xy - (1+xy) \ln(1+xy)}{y(1+xy) \ln^2 xy}$;

(4) $f'_x = -\frac{2xy}{x^4 + y^2}$, $f'_y = \frac{x^2}{x^4 + y^2}$;

(5) $f'_x = \frac{1}{\sqrt{x^2 - y^2}}$, $f'_y = \frac{\sqrt{x^2 - y^2} - x}{y\sqrt{x^2 - y^2}}$;

(6) $f'_x = y^x \ln y + yx^{y-1}$, $f'_y = x^y \ln x + xy^{x-1}$;

(7) $f'_x = -y2^{xy} \sin(1+2^{xy}) \ln 2$, $f'_y = -x2^{xy} \sin(1+2^{xy}) \ln 2$;

(8) $f'_x = \frac{|y|}{x^2 + y^2}$, $f'_y = \frac{-xy}{|y|(x^2 + y^2)}$;

(9) $f'_x = \frac{z}{y} \left(\frac{x}{y}\right)^{z-1}$, $f'_y = -\frac{zx^z}{y^{z+1}}$, $f'_z = \left(\frac{x}{y}\right)^z \ln \frac{x}{y}$;

(10) $f'_x = \begin{cases} \frac{2x((x^2 + y^2) \cos(x^2 y^2) - \sin(x^2 y^2))}{(x^2 + y^2)^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$

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(11) $f'_x = \begin{cases} 2xy^2 \arctan \frac{1}{x} \arctan \frac{1}{y} - \frac{x^2 y^2}{1+x^2} \arctan \frac{1}{y}, & x \neq 0, \\ 0, & x = 0. \end{cases}$

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2. (1)---(5)不可微, (6)可微.

3. (1) $df\left(\frac{\sqrt{2}}{2}, \frac{1}{2}, -\frac{1}{2}\right) = \left(-\frac{\sqrt{2}}{2}dx - \frac{1}{2}dy + \frac{1}{2}dz\right) \cos 1$;

$$(2) \quad df = e^{x+y+z} (dx + dy + dz) ;$$

$$(3) \quad df = \frac{xdx + ydy + zdz}{\sqrt{1+x^2+y^2+z^2}} ;$$

$$(4) \quad df = \frac{2xdx + 2ydy + 2zdz}{1+x^2+y^2+z^2} ;$$

$$(5) \quad df = \frac{2ydx - 2xdy}{(x+y)^2} ;$$

$$(6) \quad df = -\frac{e^{xy}(ydx + xdy)}{\sqrt{1-e^{2xy}}} .$$

$$4. \quad \lim_{x \rightarrow 0} \frac{f(a+x, b+x) - f(a, b)}{x} = 2 .$$

$$6. \quad (1) \text{ 近似值为 } 820\pi \text{ cm}^3. \quad (2) \quad S \approx 8.03.$$